Lab Activity 2 · Session 2 · Yash Petkar· AS25DXB021

AI Applications in Reorder Point & Inventory Management

AI in Operations (MAIB AI 218) · Session II Laboratory Exercise · S P Jain School of Global Management, Dubai · Dr. Sandip Kumar Roy

Aim. Upgrade the reorder-point calculator with (1) machine-learning demand forecasting that updates as new data arrives, (2) probabilistic and Monte Carlo safety stock, and (3) Bayesian optimisation of the reorder policy under constraints. Delivered as an executed notebook and an interactive Streamlit dashboard that share one engine.

Data. UCI *Online Retail II* (Chen, 2019; CC BY 4.0): 1,067,371 transaction lines from a UK online gift wholesaler, Dec 2009 – Dec 2011, cleaned to 1,002,005 sales lines; top 24 regular SKUs × 104 weeks. External factor: UK bank holidays. Lead time (3 ± 1 wk), costs and stock are labelled assumptions, because the data has none. Copy used: github.com/allanvc/onlineretail2, SHA-256 verified.

Key results

53.3%
Forecast: test WAPE of the auto-selected ridge regression (naive 69.8%)

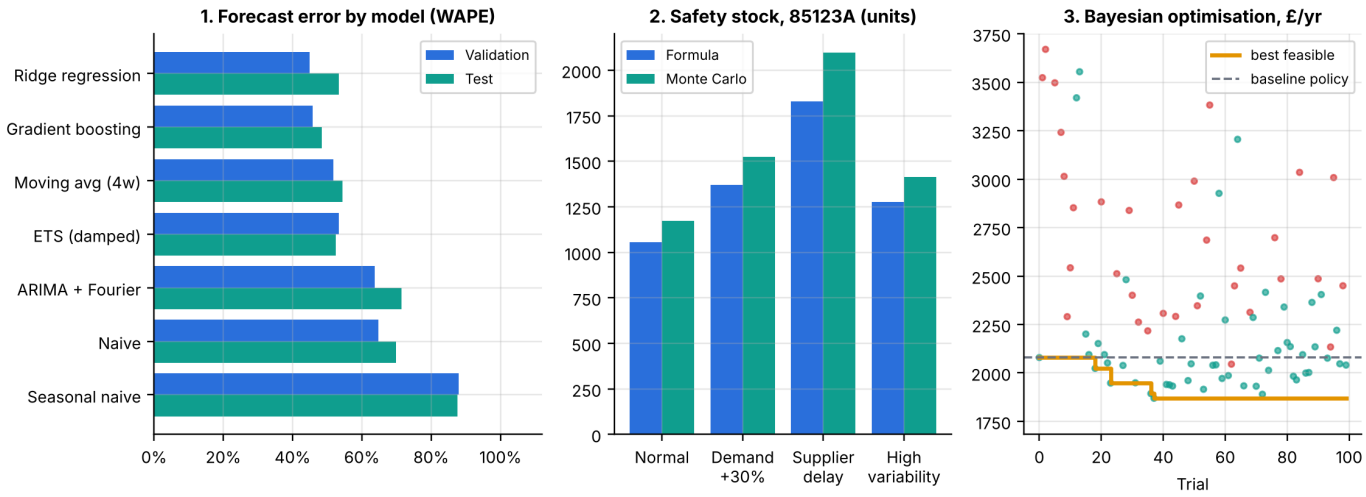
–9%
Continuous update: error when refitted weekly vs static (55.9% → 50.7%)

23/24
Monte Carlo: SKUs needing more buffer than the formula (median +13%)

20/24
Bayesian opt.: SKUs meeting 95% service (baseline 6/24), cost -6.1%

Method: 7 forecasting models (naive, seasonal naive, moving average, ETS, ARIMA + Fourier, ridge, gradient boosting) on a chronological train / validation / test split with rolling origins; SS = $Z\sqrt{(\sigma_D^2 L + D_{mean}^2 \sigma_L^2)}$ with D_{mean} = forecast mean and σ_D = forecast RMSE; 10,000-run Monte Carlo (gamma demand and lead time, copula correlation); Optuna TPE over ROP and Q with service, capacity, budget and MOQ constraints on common random numbers, validated on fresh paths.

Charts



Insights

1. **Seasonality is strong and forecastable.** November demand runs at 1.40× an average month. The selected model beats the naive forecast on unseen weeks (53.3% vs 69.8% WAPE), and refitting each week cuts error by 9% through the peak.
2. **The textbook formula under-protects skewed demand.** For 85123A the formula ROP gives 93.6% service, not 95%. A supplier delay pushes the stockout risk to 26% per cycle, and correlated delays (ρ = 0.6) raise the needed buffer 1.9× to 2,200 units.
3. **Optimising ROP and Q together pays.** For 85123A, Bayesian optimisation meets every constraint at 98.3% service and -6.8% cost. Across all 24 SKUs, service compliance rises from 6 to 20 SKUs while cost falls 6.1%.

Recommendation & conclusion

Set buffers from simulation rather than the closed-form formula, refresh forecasts weekly before the October–November peak, and let the Bayesian optimiser choose reorder point and order size per SKU within the warehouse and budget limits. All three exercise steps are implemented on real data. The limits are that lead times, costs and stock are assumed, and that refreshes are batch, not real-time.

Files: lab2_session2_yashpetkar_AS25DXB021.ipynb (executed, Colab-ready) · Streamlit dashboard: apps/streamlit/app.py